Lecture 18 Scribe

L18.pdf
PDF

"Generate a lecture scribe suitable for exam preparation based ONLY on the provided context (Lecture 18 slides). The target audience is a student preparing for a closed-notes exam.

Instructions:

Reason Step-by-Step: While writing the scribe, explicitly explain the reasoning behind definitions, assumptions, proofs, and example solutions exactly as presented in the provided context. Include this reasoning directly in the scribe text to help the student understand 'why' and 'how'.

Faithfulness: Include definitions, notation, theorem statements, and worked examples exactly as they appear in the slides. Do not hallucinate or add information from outside the context.

Logical Flow: Make the logical dependencies between concepts clear.

Structure: Organize the content with clear headings and subheadings corresponding to the lecture flow.

Goal: The output should allow a student to recall definitions, reconstruct proofs, and follow example solutions step-by-step."

Here is an **exam-oriented lecture scribe based strictly on the provided Lecture 18 slides**. I preserved the definitions, formulas, reasoning, and solved examples exactly from the slides and explained the **step-by-step reasoning behind them** so that you can reconstruct answers during a **closed-notes exam**.

Source: Lecture slides L18

Lecture 18 Scribe

Marginal PMF, Correlation, Covariance, Joint Gaussian Random Variables

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1. Marginal PMF

1.1 Definition

For discrete random variables X and Y, the **joint PMF** is defined as

$$p_{XY}(x, y) = P(X = x, Y = y)$$

The **marginal PMF** of each variable is obtained by summing the joint PMF over the other variable:

$$p_X(x) = \sum_y p_{XY}(x, y)$$

$$p_Y(y) = \sum_x p_{XY}(x, y)$$

Reasoning Behind the Definition

The joint PMF gives the probability that **both variables take specific values simultaneously**.

However, sometimes we only want the distribution of **one variable regardless of the other variable**.

To remove the effect of the other variable, we **add probabilities across all possible values of that variable**.

Key Idea:
The marginal PMF of one variable is obtained by **summing the joint PMF over all possible values of the other variable**.
L18

1.2 Marginal PMF from a Joint PMF Table

If a joint PMF is given in a table:

$p_{XY}(x, y)$	y_1	y_2	y_3	$p_X(x)$
x_1	p_{11}	p_{12}	p_{13}	$\sum_j p_{1j}$
x_2	p_{21}	p_{22}	p_{23}	$\sum_j p_{2j}$
x_3	p_{31}	p_{32}	p_{33}	$\sum_j p_{3j}$

Then:

$$p_X(x_i) = \sum_j p_{ij}$$

$$p_Y(y_j) = \sum_i p_{ij}$$

Reasoning

- **Row sums** give the marginal distribution of X
- **Column sums** give the marginal distribution of Y

This works because we are **aggregating probabilities over all possible values of the other variable**.

Also,

$$\sum_i p_X(x_i) = 1$$

$$\sum_j p_Y(y_j) = 1$$

which confirms that marginal PMFs remain valid probability distributions.

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1.3 Solved Example

Given joint PMF:

$p_{XY}(x,y)$	$y=0$	$y=1$
$x=0$	$1/8$	$1/4$
$x=1$	$3/8$	$1/4$

Step 1: Compute $p_X(x)$

Sum across **rows**.

For $x=0$

$$p_X(0) = \frac{1}{8} + \frac{1}{4}$$
$$= \frac{1}{8} + \frac{2}{8} = \frac{3}{8}$$

For $x=1$

$$p_X(1) = \frac{3}{8} + \frac{1}{4}$$
$$= \frac{3}{8} + \frac{2}{8} = \frac{5}{8}$$

Thus,

$$p_X(x) = \begin{cases} 3/8 & x=0 \\ 5/8 & x=1 \end{cases}$$

Step 2: Compute $p_Y(y)$

Sum across **columns**.

For $y=0$

$$p_Y(0) = \frac{1}{8} + \frac{3}{8} = \frac{1}{2}$$

For $y=1$

$$p_Y(1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

Thus,

$$p_Y(y) = \begin{cases} 1/2 & y = 0 \\ 1/2 & y = 1 \end{cases}$$

This shows how marginal distributions summarize the behavior of a single variable from a joint distribution.
L18

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2. Correlation

A basic way to measure the joint behavior of two random variables is through the **product moment**.

$$R_{XY} = E[XY]$$

Reasoning

This quantity measures the **average value of the product** XY.

If both variables tend to be large or small together, the product tends to be large, indicating dependence.

However:

- R_{XY} depends on the **scale of the variables**
- It is computed **about the origin, not the mean**

Therefore it mixes together:

- Mean values
- Spread of variables
- Dependence between variables

Thus it is only a **raw correlation measure**, not a clean dependence measure.
L18

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3. Covariance

To remove the effect of the means, we **center the variables**.

This leads to the definition of **covariance**.

3.1 Definition

$$Cov(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

where

$$\mu_X = E[X], \quad \mu_Y = E[Y]$$

Reasoning

Steps involved in covariance:

1. Subtract the mean from each variable.
2. Multiply the centered values.
3. Take the expected value.

This measures **how deviations from the means move together**.

Interpretation:

- $Cov(X, Y) > 0 \rightarrow$ variables increase together
- $Cov(X, Y) < 0 \rightarrow$ one increases while the other decreases
- $Cov(X, Y) = 0 \rightarrow$ no linear relationship L18

Important note:

Covariance tells the **direction** of linear variation but **not its strength**, because it depends on measurement units.

3.2 Algebraic Form of Covariance

Start with the definition:

$$Cov(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

Expand:

$$= E[XY - X\mu_Y - Y\mu_X + \mu_X\mu_Y]$$

Apply expectation linearly:

$$= E[XY] - \mu_Y E[X] - \mu_X E[Y] + \mu_X\mu_Y$$

Since

$$\mu_X = E[X], \quad \mu_Y = E[Y]$$

we obtain

$$Cov(X, Y) = E[XY] - E[X]E[Y]$$

Reasoning

This formula is easier to compute because it only requires:

- $E[X]$
- $E[Y]$

- $E[XY]$ [L18]

3.3 Properties of Covariance

1. Symmetry

$$Cov(X, Y) = Cov(Y, X)$$

Reason

Multiplication is symmetric, so the covariance expression remains unchanged.

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2. Variance as a Special Case

$$Cov(X, X) = Var(X)$$

Reason

Substituting the same variable into both arguments yields variance.

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3. Linearity with Constants

$$Cov(aX + b, cY + d) = ac\,Cov(X, Y)$$

Reason

Constants shift or scale variables but do not affect dependence structure except by scaling.

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4. Independence Implies Zero Covariance

If

$$X \perp Y$$

then

$$Cov(X, Y) = 0$$

Important:

Zero covariance means **uncorrelated**, but **does not imply independence**. [L18]

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4. Example 1 (Covariance)

Given

$$Y = -2X + 3$$

Find $Cov(X, Y)$.

Step 1: Use covariance formula

$$Cov(X, Y) = E[XY] - E[X]E[Y]$$

Step 2: Substitute the expression for Y

$$XY = X(-2X + 3)$$
$$= -2X^2 + 3X$$

Step 3: Compute expectation

$$E[XY] = E[-2X^2 + 3X]$$
$$= -2E[X^2] + 3E[X]$$

Also

$$E[Y] = E[-2X + 3]$$
$$= -2E[X] + 3$$

Step 4: Substitute into covariance formula

After simplification:

$$Cov(X, Y) = -2E[X^2] + 2(E[X])^2$$

Using

$$Var(X) = E[X^2] - (E[X])^2$$

we obtain

$$Cov(X, Y) = -2Var(X)$$

Thus,

$$Cov(X, Y) = -2Var(X)$$

This shows a **negative linear relationship** between X and Y.

L18

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5. Pearson's Correlation Coefficient

Covariance depends on measurement units.

To obtain a **unit-free measure**, we normalize covariance.

$$\rho_{XY} = \frac{Cov(X,Y)}{\sqrt{Var(X)Var(Y)}}$$

or

$$\rho_{XY} = \frac{Cov(X,Y)}{\sigma_X \sigma_Y}$$

Properties

- Measures **strength and direction** of linear relationship
- Unit-free
- Normalized version of covariance

Range:

$$-1 \leq \rho_{XY} \leq 1$$

Interpretation:

- $\rho > 0 \rightarrow$ positive linear trend
- $\rho < 0 \rightarrow$ negative linear trend
- $\rho = 0 \rightarrow$ no linear correlation

L18

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6. Jointly Gaussian Random Variables

Definition

A pair (X, Y) is **jointly Gaussian** if:

- Their joint density has the **bivariate Gaussian form**
- The marginal distributions are **Gaussian**.

Joint Gaussian PDF

$$f_{XY}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\right.\right.\right.$$

where

- μ_X, μ_Y = means
- σ_X, σ_Y = standard deviations
- ρ = Pearson correlation coefficient

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Interpretation

Example given in the slides:

Variables:

- X: Atmospheric CO_2 concentration
- Y: Global temperature anomaly

Both depend on many environmental processes.

Because of many small random effects, their joint behavior can often be approximated by a **Joint Gaussian distribution**.

If

$$\rho > 0$$

then

higher CO_2 levels tend to occur with higher temperatures. L18

Key Exam Summary

Remember these core formulas:

Marginal PMF

$$p_X(x) = \sum_y p_{XY}(x, y)$$

Covariance

$$Cov(X, Y) = E[XY] - E[X]E[Y]$$

Pearson Correlation

$$\rho_{XY} = \frac{Cov(X, Y)}{\sigma_X \sigma_Y}$$

Joint Gaussian PDF

$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} e^{(\cdots)}$$

✅ If you want, I can also make a **very short "1-page exam cheat sheet" of this lecture** that helps you solve **all possible exam questions quickly.**

Sources

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Goal: The output should allow a student to recall definitions, reconstruct proofs, and follow example solutions step-by-step."

Below is a **structured lecture scribe strictly based on Lecture 18 slides**.
All definitions, formulas, reasoning, and example steps come from the slides only so that a **student can reconstruct answers during a closed-notes exam**.

Source: Lecture slides L18

Lecture 18 Scribe

Marginal PMF, Correlation, Covariance, Joint Gaussian Random Variables

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1. Marginal PMF

1.1 Definition

For discrete random variables X and Y, the **joint probability mass function (PMF)** is

$$p_{XY}(x, y) = P(X = x, Y = y)$$

The **marginal PMFs** are obtained by summing over the other variable:

$$p_X(x) = \sum_y p_{XY}(x, y)$$

$$p_Y(y) = \sum_x p_{XY}(x, y)$$

Reasoning Behind the Definition

The joint PMF gives the probability of **both variables taking particular values simultaneously**.

However, sometimes we want the probability distribution of **only one variable** regardless of the other variable.

To remove the influence of the other variable, we **sum the joint probabilities across all possible values of that variable**.

Key Idea

The marginal PMF of one variable is obtained by **summing the joint PMF over all possible values of the other variable**.

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1.2 Marginal PMF from a Joint PMF Table

When a joint PMF is given in tabular form:

$p_{XY}(x,y)$	y_1	y_2	y_3	$p_X(x)$
x_1	p_{11}	p_{12}	p_{13}	$\sum_j p_{1j}$
x_2	p_{21}	p_{22}	p_{23}	$\sum_j p_{2j}$
x_3	p_{31}	p_{32}	p_{33}	$\sum_j p_{3j}$

Then

$$p_X(x_i) = \sum_j p_{ij}$$

$$p_Y(y_j) = \sum_i p_{ij}$$

Reasoning

- **Row sums $\rightarrow$ marginal PMF of X**
- **Column sums $\rightarrow$ marginal PMF of Y**

This works because we are **aggregating probabilities across all possible values of the other variable**.

Also the marginal distributions must satisfy

$$\sum_i p_X(x_i) = 1$$

$$\sum_j p_Y(y_j) = 1$$

which ensures they remain valid probability distributions.

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1.3 Example: Computing Marginal PMFs

The joint PMF is

$p_{XY}(x,y)$	$y=0$	$y=1$
$x=0$	$1/8$	$1/4$
$x=1$	$3/8$	$1/4$

Step 1: Compute $p_X(x)$

Add probabilities across each row.

For $x=0$

$$p_X(0) = \frac{1}{8} + \frac{1}{4}$$

Convert to common denominator:

$$\frac{1}{8} + \frac{2}{8} = \frac{3}{8}$$

For $x=1$

$$p_X(1) = \frac{3}{8} + \frac{1}{4}$$

$$\frac{3}{8} + \frac{2}{8} = \frac{5}{8}$$

Thus

$$p_X(x) = \begin{cases} 3/8 & x=0 \\ 5/8 & x=1 \end{cases}$$

Step 2: Compute $p_Y(y)$

Add probabilities across columns.

For $y=0$

$$p_Y(0) = \frac{1}{8} + \frac{3}{8}$$

$$= \frac{4}{8} = \frac{1}{2}$$

For $y=1$

$$p_Y(1) = \frac{1}{4} + \frac{1}{4}$$
$$= \frac{1}{2}$$

Thus

$$p_Y(y) = \begin{cases} 1/2 & y = 0 \\ 1/2 & y = 1 \end{cases}$$

This demonstrates how marginal PMFs summarize the behavior of each variable independently. L18

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2. Correlation

A basic way to measure the joint behavior of two random variables is the **product moment**.

$$R_{XY} = E[XY]$$

Reasoning

This quantity represents the **average value of the product** XY.

However:

- R_{XY} depends on the **scale of** X **and** Y.
- It is computed **about the origin rather than about the means**.

Because of this, it mixes together

- Mean values
- Spread of the variables
- Dependence between variables

Thus it is **only a starting point for measuring dependence**.

If

$$R_{XY} = 0$$

then the variables are **orthogonal about the origin**.

To remove the influence of the means, we **center the variables**, which leads to **covariance**. L18

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3. Covariance

3.1 Definition and Interpretation

Covariance is defined as

$$Cov(X,Y) = E[(X-\mu_X)(Y-\mu_Y)]$$

where

$$\mu_X = E[X], \quad \mu_Y = E[Y]$$

Reasoning

The procedure for computing covariance is:

1. Subtract the mean from each variable.
2. Multiply the centered values.
3. Take the expectation.

This measures **how deviations from the means vary together**.

Interpretation:

- $Cov(X,Y) > 0$
  $\rightarrow$ positive covariance (variables tend to increase together)
- $Cov(X,Y) < 0$
  $\rightarrow$ negative covariance (one increases when the other decreases)
- $Cov(X,Y) = 0$
  $\rightarrow$ zero covariance (no linear co-variation trend)

Covariance tells the **direction of joint linear variation**, but its magnitude depends on units, so it cannot measure strength reliably.

L18

3.2 Algebraic Form of Covariance

Start with the definition

$$Cov(X,Y) = E[(X-\mu_X)(Y-\mu_Y)]$$

Expand the product

$$= E[XY - X\mu_Y - Y\mu_X + \mu_X\mu_Y]$$

Apply linearity of expectation

$$= E[XY] - \mu_Y E[X] - \mu_X E[Y] + \mu_X\mu_Y$$

Using

$$\mu_X = E[X], \quad \mu_Y = E[Y]$$

we obtain

$$Cov(X,Y) = E[XY] - \mu_X\mu_Y$$

or

$$Cov(X,Y) = E[XY] - E[X]E[Y]$$

Reasoning

This algebraic form simplifies computation because it only requires calculating:

- $E[X]$
- $E[Y]$
- $E[XY]$

rather than working directly with centered variables.
L18

3.3 Properties of Covariance

1. Symmetry

$$Cov(X,Y) = Cov(Y,X)$$

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2. Variance as a Special Case

$$Cov(X,X) = Var(X)$$

---

3. Linearity with Respect to Constants

$$Cov(aX+b, cY+d) = ac\,Cov(X,Y)$$

---

4. Independence Implies Zero Covariance

If

$$X \perp Y$$

then

$$Cov(X,Y) = 0$$

Important note:

Zero covariance means the variables are **uncorrelated**, but **uncorrelated does not necessarily imply independence**.
L18

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4. Example 1: Covariance

Given

$$Y = -2X + 3$$

Find $Cov(X, Y)$.

Step 1: Start with the covariance formula

$$Cov(X, Y) = E[XY] - E[X]E[Y]$$

Step 2: Substitute the expression for Y

$$XY = X(-2X + 3)$$
$$= -2X^2 + 3X$$

Step 3: Compute expectations

$$E[XY] = E[-2X^2 + 3X]$$
$$= -2E[X^2] + 3E[X]$$

Also

$$E[Y] = E[-2X + 3]$$
$$= -2E[X] + 3$$

Step 4: Substitute into covariance formula

After simplifying,

$$Cov(X, Y) = -2E[X^2] + 2(E[X])^2$$

Using the variance identity

$$Var(X) = E[X^2] - (E[X])^2$$

we obtain

$$Cov(X, Y) = -2Var(X)$$

Thus

$$Cov(X, Y) = -2Var(X)$$

This indicates a **negative relationship between X and Y**.

L18

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5. Pearson's Correlation Coefficient

Covariance depends on measurement units.
To remove unit dependence, we normalize covariance.

$$\rho_{XY} = \frac{Cov(X,Y)}{\sqrt{Var(X)Var(Y)}}$$

or

$$\rho_{XY} = \frac{Cov(X,Y)}{\sigma_X \sigma_Y}$$

Properties

- Measures **strength and direction of linear relationship**
- Normalized version of covariance
- Unit-free

Range

$$-1 \leq \rho_{XY} \leq 1$$

Interpretation

- $\rho_{XY} > 0 \rightarrow$ positive linear trend
- $\rho_{XY} < 0 \rightarrow$ negative linear trend
- $\rho_{XY} = 0 \rightarrow$ no linear correlation

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6. Jointly Gaussian Random Variables

Definition

A pair (X, Y) is **jointly Gaussian** if

- its joint density has the **bivariate Gaussian form**, and
- the marginal distributions are **Gaussian**.

L18

6.1 Joint Gaussian PDF

For two variables X and Y

$$f_{XY}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y}{}\right.\right.\right.$$

where

- μ_X, μ_Y = means

- σ_X, σ_Y = standard deviations
- ρ = Pearson correlation coefficient

6.2 Interpretation and Example

Example given in the lecture:

Variables studied:

- X: Atmospheric CO_2 concentration (ppm)
- Y: Global surface temperature anomaly (°C)

Because these variables depend on many environmental factors (solar radiation, ocean circulation, emissions, weather variability), their joint behavior can often be approximated by a **joint Gaussian distribution**.

If

$$\rho > 0$$

then

higher CO_2 levels tend to occur with higher temperatures.
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End of Lecture 18 Scribe

This structure allows a student to:

- Recall **definitions directly**
- Reconstruct **derivations step-by-step**
- Follow **worked examples exactly as shown in the lecture**.

Sources

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Below is a **faithful lecture scribe for Lecture 18**, written strictly from the provided slides.
The goal is **closed-notes exam preparation**, so the scribe explains **definitions, derivations, and examples step-by-step exactly as shown in the slides** while clarifying the reasoning behind each step.

Source: Lecture slides L18

Lecture 18 Scribe

Marginal PMF, Correlation, Covariance, Joint Gaussian Random Variables

---

1. Marginal PMF

1.1 Definition

For **discrete random variables** X and Y, the **joint PMF** is defined as

$$p_{XY}(x, y) = P(X = x, Y = y)$$

This function gives the probability that **both variables simultaneously take particular values**.

The **marginal PMF** of each variable is obtained by summing over the other variable:

$$p_X(x) = \sum_y p_{XY}(x, y)$$

$$p_Y(y) = \sum_x p_{XY}(x, y)$$

Reasoning Behind the Definition

The joint PMF describes the **combined behavior of two variables**.

However, sometimes we want to know the distribution of **only one variable** regardless of the value of the other variable.

To obtain this, we **sum all probabilities involving that value of the variable across every possible value of the other variable**.

Key idea from the slides:

The marginal PMF of one variable is obtained by summing the joint PMF over all possible values of the other variable. L18

1.2 Marginal PMF from a Joint PMF Table

When the joint PMF is given in a table, marginal PMFs can be computed by summing rows and columns.

$p_{XY}(x, y)$	y_1	y_2	y_3	$p_X(x)$
x_1	p_{11}	p_{12}	p_{13}	$\sum_j p_{1j}$
x_2	p_{21}	p_{22}	p_{23}	$\sum_j p_{2j}$
x_3	p_{31}	p_{32}	p_{33}	$\sum_j p_{3j}$

From this structure:

$$p_X(x_i) = \sum_j p_{ij}$$

$$p_Y(y_j) = \sum_i p_{ij}$$

Reasoning

- **Row sums $\rightarrow$ marginal distribution of X**
- **Column sums $\rightarrow$ marginal distribution of Y**

This procedure works because the probability of $X = x_i$ includes **all outcomes where $X = x_i$ occurs regardless of Y**.

Also, the marginal PMFs must satisfy the normalization property

$$\sum_i p_X(x_i) = 1$$

$$\sum_j p_Y(y_j) = 1$$

which ensures they remain valid probability distributions. L18

1.3 Example: Computing Marginal PMFs

Given joint PMF:

$p_{XY}(x, y)$	$y = 0$	$y = 1$
$x = 0$	$1/8$	$1/4$
$x = 1$	$3/8$	$1/4$

Step 1: Compute $p_X(x)$

Sum the probabilities across each row.

For $x = 0$:

$$p_X(0) = \frac{1}{8} + \frac{1}{4}$$

Convert to common denominator:

$$\frac{1}{8} + \frac{2}{8} = \frac{3}{8}$$

For $x = 1$:

$$p_X(1) = \frac{3}{8} + \frac{1}{4}$$

$$\frac{3}{8} + \frac{2}{8} = \frac{5}{8}$$

Thus

$$p_X(x) = \begin{cases} 3/8 & x = 0 \\ 5/8 & x = 1 \end{cases}$$

Step 2: Compute $p_Y(y)$

Sum the probabilities across columns.

For $y = 0$:

$$p_Y(0) = \frac{1}{8} + \frac{3}{8} = \frac{4}{8} = \frac{1}{2}$$

For $y = 1$:

$$p_Y(1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

Thus

$$p_Y(y) = \begin{cases} 1/2 & y = 0 \\ 1/2 & y = 1 \end{cases}$$

This example illustrates how marginal PMFs are computed by **aggregating probabilities from the joint distribution**.

L18

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2. Correlation

A basic way to measure the joint behavior of two random variables is through the **product moment**

$$R_{XY} = E[XY]$$

Reasoning

This quantity represents the **average value of the product** XY.

However, the slides explain several limitations:

- R_{XY} depends on the **scale of** X **and** Y.
- It is computed **about the origin, not about the means**.

Because of this, the value mixes together:

- Mean values
- Spread of the variables
- Dependence between variables

Thus it is **not a clean measure of centered dependence**.

If

$$R_{XY} = 0$$

then the variables are **orthogonal about the origin**.

To obtain a better dependence measure, we **remove the effect of the means** by centering the variables.
This leads to **covariance**.
L18

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3. Covariance

3.1 Definition and Interpretation

Covariance is defined as

$$Cov(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

where

$$\mu_X = E[X], \quad \mu_Y = E[Y]$$

Reasoning

The idea is to analyze how **deviations from the mean occur together**.

Steps involved:

1. Subtract the mean from each variable.
2. Multiply the deviations.
3. Take the expectation.

Interpretation:

- $Cov(X, Y) > 0$: positive covariance
  (deviations tend to have the same sign)
- $Cov(X, Y) < 0$: negative covariance
  (one variable increases while the other decreases)
- $Cov(X, Y) = 0$: zero covariance
  (no linear co-variation trend)

Covariance shows the **direction of joint linear variation**, but the magnitude depends on measurement units.
L18

3.2 Algebraic Form of Covariance

Start with the definition

$$Cov(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

Expand the product

$$= E[XY - X\mu_Y - Y\mu_X + \mu_X\mu_Y]$$

Apply linearity of expectation

$$= E[XY] - \mu_Y E[X] - \mu_X E[Y] + \mu_X\mu_Y$$

Since

$$\mu_X = E[X], \quad \mu_Y = E[Y]$$

we obtain

$$Cov(X, Y) = E[XY] - \mu_X\mu_Y$$

or

$$Cov(X, Y) = E[XY] - E[X]E[Y]$$

Reasoning

This expression is easier to compute because we only need:

- $E[XY]$
- $E[X]$
- $E[Y]$

instead of working with centered variables directly.
L18

3.3 Properties of Covariance

1. Symmetry

$$Cov(X, Y) = Cov(Y, X)$$

---

2. Variance as a Special Case

$$Cov(X, X) = Var(X)$$

---

3. Linearity with Respect to Constants

$$Cov(aX + b, cY + d) = ac\,Cov(X, Y)$$

---

4. Independence Implies Zero Covariance

If

$$X \perp Y$$

then

$$Cov(X, Y) = 0$$

Important note from the slides:

Zero covariance means uncorrelated, but it does not imply independence. L18

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4. Example 1: Covariance

Given

$$Y = -2X + 3$$

Find $Cov(X, Y)$.

Step 1: Use the covariance formula

$$Cov(X, Y) = E[XY] - E[X]E[Y]$$

Step 2: Substitute the expression for Y

$$XY = X(-2X + 3)$$

$$= -2X^2 + 3X$$

Step 3: Compute expectations

$$E[XY] = E[-2X^2 + 3X]$$
$$= -2E[X^2] + 3E[X]$$

Also

$$E[Y] = E[-2X + 3]$$
$$= -2E[X] + 3$$

Step 4: Substitute into covariance formula

After simplification:

$$Cov(X, Y) = -2E[X^2] + 2(E[X])^2$$

Using the variance identity

$$Var(X) = E[X^2] - (E[X])^2$$

we obtain

$$Cov(X, Y) = -2Var(X)$$

Thus

$$Cov(X, Y) = -2Var(X)$$

This indicates a **negative relationship between X and Y**. L18

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5. Pearson's Correlation Coefficient

Covariance depends on the units of measurement.

To obtain a **unit-free measure**, we normalize covariance.

$$\rho_{XY} = \frac{Cov(X, Y)}{\sqrt{Var(X)Var(Y)}}$$

or

$$\rho_{XY} = \frac{Cov(X, Y)}{\sigma_X \sigma_Y}$$

Properties:

- Measures **strength and direction of linear relationship**
- Normalized version of covariance

- Unit-free

Range:

$$-1 \leq \rho_{XY} \leq 1$$

Interpretation:

- $\rho_{XY} > 0 \rightarrow$ positive linear trend
- $\rho_{XY} < 0 \rightarrow$ negative linear trend
- $\rho_{XY} = 0 \rightarrow$ no linear correlation L18

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6. Jointly Gaussian Random Variables

Definition

A pair (X, Y) is **jointly Gaussian** if

- its joint density has the **bivariate Gaussian form**, and
- the marginal distributions are **Gaussian**. L18

6.1 Joint Gaussian PDF

For variables X and Y

$$f_{XY}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\right.\right.\right.$$

where

- μ_X, μ_Y are the means
- σ_X, σ_Y are the standard deviations
- ρ is the Pearson correlation coefficient

6.2 Interpretation Example

Example from the slides:

Variables:

- X: Atmospheric CO_2 concentration (ppm)
- Y: Global surface temperature anomaly (°C)

Because both variables vary due to many environmental processes (solar radiation, ocean circulation, emissions, weather variability), their joint behavior can often be approximated by a **joint Gaussian**

distribution.

If

$$\rho > 0$$

then **higher CO_2 levels tend to occur with higher temperatures**.

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✔️ This lecture scribe follows the **exact lecture flow** so a student can:

- Recall **definitions**
- Reconstruct **derivations**
- Solve **examples step-by-step during an exam**.

Sources